

CHAPTER 7

Wages and Employment in a Single Labour Market

Answers to End-of-Chapter Questions

1. (LO2) Each demand curve would become steeper and in most cases, less elastic. For these individual firms, nothing would happen to the wage levels in the short run due to the fact that they face an infinitely elastic labour supply schedule. This means that the wage is supply-determined. For the entire labour market, one would expect a rotation of the labour demand curve that would bring about a fall in demand, and hence a decrease in the equilibrium market wage and in the equilibrium employment level. Since the labour supply curve facing each individual firm is perfectly elastic at the going market wage, each labour supply curve would thus decrease at the end of the exercise. The equilibrium market wage would decline, and thus the wage paid by each firm would decline.
2. (LO2, 3) Figure 7.6 essentially illustrates the situation. The labour demand curve depicted in this figure would change, however. It is no longer the value of the marginal product curve, but rather the marginal revenue product curve: labour demand = $MRP_N = MPP_N \cdot MR$. The labour demand curve becomes steeper and less elastic. Upon imposing a labour supply curve, one can expect a lower level of employment and wage than appears in Figure 7.6. Is it necessary that monopoly be accompanied by monopsony? Not at all. Is it possible that the two conditions could go together? Yes, but in the real world, monopsony is a fairly uncommon situation, limited to a few occupations in fairly lightly populated regions.

For question 3, refer to Figure 7.6 of the textbook.

3. (LO3) The supply curve that we see in Figure 7.6 is pretty elastic. A less elastic supply curve cutting through S_m would be steeper. The latter supply curve would have a marginal cost of labour curve that is steeper and higher than the MC curve in Figure 7.6. This would generate an equilibrium wage–employment combination on the labour supply curve which lies below S_m . The less elastic supply curve generates an equilibrium which is less favourable to the labour force. This result makes intuitive sense, as less elastically supplied labour is less responsive to wage changes (less likely to quit when the pay is cut, less likely to work more when the pay is raised) than is the case for elastically supplied labour.
4. (LO3) Yes, it is possible for a multi-million dollar professional sports player to be subject to monopsonistic exploitation of labour. Imagine that the player can sell her services only to the team which signed her, which was the case in baseball until the late 1970s. Other than accepting the employer's offer, her only option is to quit. Assuming that she does have some outside options which have reservation wages attached to them, the labour supply curve will slope upward, and the equilibrium can be modelled as in Figure 7.6. She will earn far more than her reservation wage (what she would earn in the alternative occupation) – even millions, perhaps, but the compensation will be less than it would be

in the competitive labour market (one in which she could sell her services to the highest bidder among other sports teams).

Is it possible for workers who are subject to monopsonistic exploitation of labour to be receiving an economic rent on the sale of their labour market services? Yes. In Figure 7.6, the only worker who is not receiving an economic rent (a wage higher than his or her reservation wage) is the last worker hired at point N_M . . Could workers who are receiving an economic rent for their services ever be considered disadvantaged workers? The term “disadvantaged” is quite subjective. Most workers probably consider themselves to be underpaid, while much of the rest of the labour force would beg to differ. Any worker will be paid at least as much as their reservation wage, but this can be a very low wage, from which the employer is extracting surplus (the lion's share of the marginal revenue product). In the theory of monopsony, all workers receive a wage which is lower than the wage that they could receive if the labour market were competitive. One might say that they are disadvantaged by the monopsonist, who is exercising power in the labour market and not paying them the value of their contribution to the firm's revenues.

5. (LO4) Explain why a minimum wage increase, over a certain range, would lead to a monopsonist actually increasing employment. See the subsection entitled “Expected Impact: Monopsony in the Labour Market” of the text.

Given this possibility, could the monopsony argument be relied upon to argue against the critics of minimum wage legislation who argue that minimum wages will have an adverse employment effect and hence will harm some of the very people they were designed to help? Yes, this model does counter the frequent criticism that the minimum wage has an adverse employment effect. Note that it is only applicable for a certain range of the minimum wage, however, and it only applies in the case where the employer has monopsony power. If the minimum wage were to be raised substantially, it probably would cause an adverse employment effect, even if the employer were a monopsonist. This would be the case if the minimum wage were to be raised above the wage that would prevail in a competitive labour market (the intersection of VMP and $S = AC$ in Figure 7.9 of the text).

Could minimum wages ever be applied selectively to monopsony situations? Could wage fixing via unionization be applied more selectively? Minimum wages and union wage setting work in the same fashion. The answer is essentially given in the question above. The imposed wage has to lie between the wage that the monopsonist would set on its own and the wage that would prevail in the absence of market power (i.e., at the intersection of the supply and VMP schedule at N_1). In that subsection of the text, the point is made that the sectors which are most likely to pay minimum wages are not those which are likely to have monopsony features. Minimum wages tend to be paid in labour markets which are competitive in nature.

For question 6, refer to Figure 7.9 of the textbook.

6. (LO4) The closer the labour market equilibrium is to the competitive level (given by the intersection of the VMP and supply of labour curves), the more efficient is the allocation of labour. The value of production increases by region under the VMP curve between N_0 and N_1 . The opportunity cost of this employment increase, as measured by the height of the labour supply curve, increases by the area under the supply curve between N_0 and N_1 . This quantity can be considered the cost of employing these extra labour resources in terms of leisure (or possibly work with other employers) foregone. Since the marginal benefit of this employment gain outweighs the marginal cost, there is a net gain in efficiency.

If the monopsonist followed a policy of perfect wage differentiation in the absence of a minimum wage, what would be the implications for resource allocation? Compare the income distribution consequences of the two alternatives of minimum wages and perfect wage differentiation: The implications for resource allocation would be the same in either case, as the wage/employment equilibrium is identical. The workers would be paid a lot less, but the profits would be a lot higher, however, under the latter scheme. Thus the distributional implications are not the same at all, as the workers would be much worse off, and the firm would be much better off, in the case of perfect wage discrimination.

7. (LO1, 2)

This statement is true. The scenario described is totally analogous to the imposition of a payroll tax, except that in this case of a subsidy, the tax is negative. Consider the case of a wage subsidy paid that is paid to the firm. This will cause an increase in the demand for labour, and the demand curve will shift to the right (upward by the amount of the tax). At each possible employment level, firms are willing to pay a higher wage, as part of the wage is footed by the government. Alternatively, at each possible wage level, the quantity of labour demanded is higher. All other factors held constant, the equilibrium wage paid in the market (net of the subsidy) will rise, as will the equilibrium level of employment.

Consider next the case of a wage subsidy that is paid to the worker. This will cause an increase in the supply of labour, and the supply curve will shift to the right. This shift will be of the same magnitude as in the previous case. At each possible employment level, workers are willing to accept a lower wage rate, as they are receiving part of their pay from the government. Alternatively, at each possible wage level, the quantity of labour supplied is higher. All other factors held constant, the equilibrium wage paid in the market (net of the subsidy) will fall, and the equilibrium level of employment. The wage plus the subsidy that the workers receive in the latter case will equal the wage that they are paid in the former case.

Answers to End-of-Chapter Problems

1. (LO4, 5) This statement is true if the prevailing market wage is above the wage that would prevail in a competitive equilibrium. In a competitive equilibrium, there is no

monopsonistic power in the labour market, and the equilibrium wage is determined by the intersection of total labour demand and total labour supply. On the other hand, if the prevailing market wage is below the wage that would prevail in a competitive equilibrium due to the exercise of monopsony power, then the statement is false. This is explained thoroughly in the subsection entitled “Expected Impact: Monopsony in the Labour Market” on page 211 of the text. There is an intuitive explanation, a formal exposition with graphical analysis, and an example with figures. Note that this somewhat tricky result – that the implementation of a minimum wage can actually raise the level of employment given a monopsony – applies only to a particular range of wages. This particular range is between the monopsony wage and the competitive equilibrium wage. Since one would never observe a wage lower than a monopsony wage (except for the case of perfect wage differentiation), the case of a minimum wage below the monopsony wage is irrelevant.

2. (LO3) We are told that stars are in short supply, so the supply curve for them is probably steep and inelastic. If the supply of grinders is unlimited, then the supply curve is highly elastic, and in the extreme is perfectly inelastic.
 - a) This situation is a monopsony. In the case of the stars, for which the supply curve is upward sloping, the team can exploit some monopsony power, and thus pay wages to the stars that are below competitive levels. The marginal labour cost curve lies above the supply of labour curve. The intersection of the marginal labour cost curve and the demand for labour curve gives the equilibrium level of employment. The wage is determined from the labour supply curve point that is consistent with that level of employment. In the case of the non-stars, if the supply is perfectly elastic, the monopsony equilibrium is no different from the competitive one, as the marginal labour cost curve coincides with the supply of labour curve. In this case, the wage differential between the stars and the non-stars is lower than it would be in the absence of monopsony power.
 - b) Given the institution of free agency, the signing team no longer has monopsony power. When the contracts expire, the labour market becomes competitive. For both the stars and the non-stars, the equilibrium wage is determined by the intersection of labour demand and labour supply. It is under these conditions that we would expect the differential between the two grades of players to be the widest.
 - c) This case is the intermediate case between cases (a) and (b). Because there is free agency, there is no monopsony power. This means that the wages will be determined by the intersection of labour demand and labour supply. The fact that revenue is shared, however, is likely to reduce the demand for stars to a level that is lower than what would otherwise be the case, because teams will no longer have the means to outbid other teams. In this case, the wage differential between the stars and the non-stars is lower than it would be in the case of an unfettered free market with free agency.

3. (LO1, 2) This problem is somewhat hard and lengthy to answer, but it does illustrate some very useful tools and concepts. You should try to work through it.

- a) If the value of a is negative, then there will be an inverse (negative) relationship between L^D and W . This result can be demonstrated by taking the derivative of the labour demand function with respect to W , which yields $AaW^{(a-1)}$. As long as a is negative, the wage variable will go into the denominator, but it will still be positive. We then have a positive value multiplied by a , which will yield a negative value for the derivative (slope) of the labour demand function. If the value of b is positive, then there will be a direct (positive) relationship between L^S and W . This result can be demonstrated by taking the derivative of the labour supply function with respect to W , which yields $BbW^{(b-1)}$. As long as the value of b is positive, the wage variable could go into the denominator or the numerator; either way it will still be positive. We then have a positive value multiplied by b , which will yield a positive value for the derivative (slope) of the labour supply function. It is harder to show these results without using calculus, so please see the following example.

If a is equal to -1 , for example, then $L^D = -A/W$, and we have a negative relationship between L^D and W . If b is equal to $+1$, for example, then $L^S = BW$, and we have a positive relationship between L^S and W . The labour supply curve would be a linear function passing through the origin with a positive slope of B (assume that $B > 0$). The labour demand curve would be the reciprocal function, having the 2 axes, wage and labour quantity, as asymptotes. The labour demand has a negative slope, assuming that A is positive. More generally, the equilibrium condition is $Aw^a = Bw^b$, which solves algebraically to $w = (A/B)^{1/(b-a)}$.

- b) Take the natural logarithm of both sides of the equations, using the basic axioms of logarithms (that the log of a product is the sum of the logs, and that the log of an exponential operation is the exponent multiplied by the base). Plotting the log of the wage on the vertical axis and the log of the employment level on the horizontal axis, you should have a linear labour demand function with an intercept of A' , a (negative) slope of a , a linear labour supply function with an intercept of B' , and a (positive) slope of b . In order to solve for the equilibrium log wage level, set $l^D = l^S$, and solve for w . You should obtain $w = (B' - A')/(a - b)$. Substitute this value for w into either the original l^D or l^S equation in order to obtain $l = (B'a - A'b) / (a - b)$. As the textbook mentions, a is the wage elasticity of demand, as it is the derivative of the log of the labour demand with respect to the log of the wage, which is equivalent to the percentage change in labour demand divided by the percentage change in wage. b is the wage elasticity of supply, as it is the derivative of the log of the labour supply with respect to the log of the wage, which is equivalent to the percentage change in labour supply divided by the percentage change in the wage.

- c) Assuming that the tax is collected from firms, only the labour demand function changes. Replacing $\ln W$ with $\ln W(1 + t)$, we obtain $\ln W + \ln(1 + t) = \ln W + t$. (This approximation is only valid for small values of t .) The new labour demand function reads $l^D = A' + aw + at$. To determine equilibrium, set $l^D = l^S = A' + aw + at = B' + bw$. The equilibrium wage is $w = (B' - A' - at)/(a - b)$. This is the market wage as well as the take-home wage. Notice how it is lower than it was in part (b). The firm pays this wage to its workers, plus it remits t to the government. The employment level is $(B'a - A'b - at^2)/(a - b)$. Notice how it is lower than it was in part (b). The labour elasticity is denoted by a , and the supply elasticity is denoted by b . As the value of a increases, the equilibrium wage falls relative to the case of no tax (in the formula above, the numerator falls and the denominator rises), which implies that the share paid by the workers rises. This makes sense because the demand for labour is becoming more elastic. The effect of an increase in the value of b is to decrease the denominator and thus to increase the ratio relative to the case with no tax, which implies that the share paid by the workers falls. This makes sense because the supply of labour is becoming more elastic. If we set $a = 0$, the labour demand is totally inelastic, and the firms pay all of the tax as the market wage is unaffected, and firms still have to pay the tax. If we set $b = 0$, the labour supply is totally inelastic, and the workers pay all of the tax as their wages are docked by the full amount of t . In the long run we expect the value of b to be quite low, implying that workers will bear an increasing share of the tax.
- d) Assuming that the tax is collected from workers, only the labour supply function changes. Replacing $\ln W$ with $\ln W(1 - t)$, we obtain $\ln W + \ln(1 - t) = \ln W - t$. (This approximation is only valid for small values of t .) The new labour supply function reads $l^S = B' + bw - bt$. To determine equilibrium, set $l^D = l^S = A' + aw = B' + bw - bt$. The equilibrium wage is $w = (B' - A' - bt)/(a - b)$. This is the market wage, but it is not the take-home wage, because the workers have not paid their taxes yet. To obtain the take-home wage, subtract t from the expression, and you obtain $(B' - A' - at)/(a - b)$, which is the same take-home pay as in part (c). Since we have demonstrated that the wage is the same, then it follows that the employment level is the same.
4. (LO3) This statement should be qualified. It is true that if the monopsonist wants to hire another worker, it has to move up its upward-sloping labour supply curve a little bit, which means that the wage paid to the entire work force increases. In comparison, the employer in a competitive labour market can recruit another employee by paying her the same wage as all of the other workers are receiving. The cost of hiring the marginal worker can be higher in monopsony. The incorrect part of the statement is that the monopsonist operates at a disadvantage due to the differing nature of labour supply. All other factors held constant, the total wage bill is lower in a monopsonistic setting. The monopsonist will pay a lower equilibrium wage and employ a lower number of people than is the case in the competitive labour market. This is illustrated in Figure 7.6 of the text.

5. (LO1, 2) This problem is not very difficult, but it is easy to make a mistake. You should ask yourself whether your answer makes economic sense or not. Like problem 3, it illustrates some good concepts and tools.
- When the firm maximizes its profits, it sets $VMP_L = \text{wage}$, so the labour demand function becomes $W = 30 - 2L$. If you are ever in doubt as to which substitution to make, draw a quick sketch of your demand or supply curve in order to see if the slope is correct. Since this is a labour demand function, the slope should be negative.
 - Rewrite the labour demand function as $L = 15 - W/2$. Since this firm-level function must be added for 19 firms, one can multiply by 10 as the firms are all identical. Recall that the market demand curve is obtained by summing the individual labour demand curves horizontally. This means that we multiply the abscissa (the L variable) by 10 to obtain the market demand curve $L = 150 - 5W$. Think of blowing up the x -axis by a factor of 10 but not changing the scale of the y -axis at all. That step is a bit tricky. To find the equilibrium, set labour supply = labour demand in order to obtain $W = 10$ and $L^D = L^S = 100$.
 - The labour demand curve shifts upward in parallel fashion by \$3. Take the market labour demand curve, and rewrite it as $W = 30 - L^D / 5$. When it shifts upward by \$3, the slope remains the same, but the intercept increases to 33, so the equation for the labour demand curve becomes $W = 33 - L^D/5$, or $L^D = 165 - 5W$. Think of it in this fashion: at each level of employment L , firms are willing to pay a wage that is \$3 higher because they are collecting a subsidy. Setting this L^D equal to L^S gives an equilibrium wage of \$11 and an equilibrium employment level of 110. The workers definitely benefit.
 - This would have the effect of shifting the labour supply curve down and to the right by \$3. The original $L^S = 10W$, so solve for W in order to obtain $L^S/10$. At each possible employment level, workers are willing to work for \$3 less per hour because they are receiving the subsidy. The equation for labour supply becomes $W = -3 + L^S/10$. Set this labour supply function equal to the labour demand function and solve the system. The equilibrium wage is now \$8, but once workers collect their \$3 bonus from the government, their take-home wage becomes \$11, just like in case (c). The equilibrium level of employment is 110. The opposition party is wrong, as the impact of the two policy measures on the labour market equilibrium is the same.
6. (LO3) No, there is not clear evidence of monopsony power at work. The proposition is assuming that doctors are more mobile than business owners, rendering the wage elasticity of supply higher for doctors than for business owners. Assuming that the labour supply decisions of the husband and wife are made jointly, this might imply that the labour supply of the professor who is married to the doctor is more elastic than the labour supply of the professor who is married to the business owner. Under these circumstances,

it would definitely be more feasible for the employer to exert monopsony power in the case of the latter. In order to claim that the employer is doing so, however, one has to establish that the demand conditions are identical. The demand for the services of each professor should be commensurate with their marginal revenue products.

7. (LO2) This statement is true. As mentioned in the helpful hints section in *istudy*, there is not enough information to determine labour market outcomes in this statement. The subsection entitled “Product Market Structures and Departures from Market Wages” of the text deals with the ambiguous question of the link between the equilibrium wage level and the degree of competition in the output market. In general, we need information on demand and supply conditions in the labour market. In practice, it seems as though employers having market power in the output market do tend to pay a wage premium, but there is no theoretical reason why this has to be so.
8. (LO1, 2) This is an interesting economic and policy issue, as payroll taxes are a favourite way for governments to raise revenue for popular social programs. Payroll taxes are actually lower in Canada than they are in the US and in many European countries. The taxes certainly have an influence on labour demand, but what about the flipside of the taxes, notably the benefits? They have an effect on labour supply. See Figure 7.5 of the text.
 - a) The answer to this question is almost given in the preamble to the question itself. The labour supply curve N^s shifts downward vertically by the distance of G . At each possible level of employment, workers require a wage level of $W - G$ instead of W in order to offer that quantity of labour. This constitutes an expansion in labour supply.
 - b) If $G = T$, the magnitude of the downward shift in the labour demand curve equals the downward shift in the labour supply curve. Compared with the original equilibrium at point A, we will have the same equilibrium level of employment (N_0) with an equilibrium wage that is lower by the magnitude $T = G$. Compared with the case in which workers do not value the benefit at all and employers are assessed the tax (point B), we arrive at a higher level of employment and a lower wage when workers do value the benefit. If workers only value the benefit somewhat (i.e., $G < T$), the equilibrium would be characterized by lower employment and a lower wage compared to the original equilibrium at point A. Therefore, the impact of workers valuing the benefit is to partially or fully mitigate the adverse impact of the tax on equilibrium employment. There is no way, however, to mitigate the adverse effect on wages. Whether or not workers value the benefit, wages will always be lower, and thus the pensions cannot ever be “free.” This analysis implies that payroll taxes imposed to finance social benefits that are valued by workers are not quite the “job killers” that they are made out to be by certain observers.